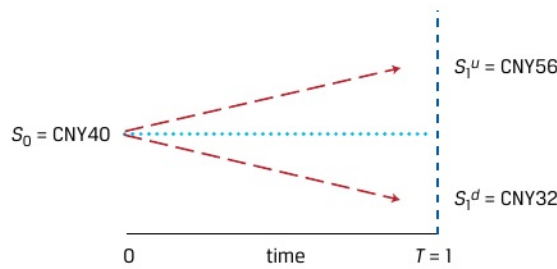
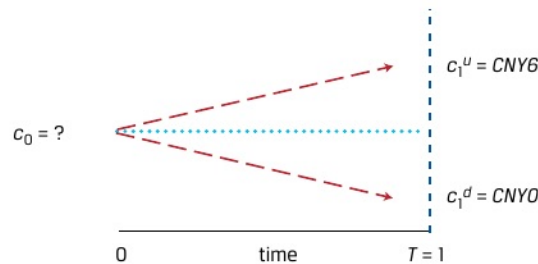


Exhibit 14: One-Period Binomial Tree for the Asset's Price

The contract value under the two scenarios is as follows (shown visually in Exhibit 15):

- Price Increase: The 40 percent increase results in a contract value of CNY 6 ($=\text{CNY}56 - \text{CNY}50$), as the contract owner will choose to buy the asset at CNY50 and is able to sell it at a market price of CNY56.
- Price Decrease: The 20 percent decrease results in a contract value of zero (CNY0), as the contract owner will choose not to buy the asset at CNY50 when the market price is only CNY32.

Exhibit 15: One-Period Binomial Tree for the Contract's Price

Consider the position of the contract seller as follows:

- At $t = 0$, the contract value is c_0 .
- At $t = 1$, the contract value is either CNY6 (if the underlying asset rises to CNY56) or CNY0 (if the underlying stock falls to CNY32).

The initial contract value c_0 is unknown and may be determined using cash flow additivity and no-arbitrage pricing. That is, the value of the contract and the underlying asset in each future scenario may be used to construct a risk-free portfolio, or a portfolio where the value is the same in both scenarios. For example, assume at $t = 0$ an investor creates a portfolio in which the contract is sold at a price of c_0 and 0.25 units of the underlying asset are purchased. This portfolio is called a replicating portfolio in that it is designed specifically to create a matching future cash flow stream to that of a risk-free asset. Denoting the value of the portfolio as V at $t = 0$ and at $t = 1$ under both the price increase and price decrease scenarios, we have the following:

- $V_0 = 0.25 \times 40 - c_0$,
- $V_1^u = 0.25 \times 56 - 6 = 8$, and
- $V_1^d = 0.25 \times 32 - 0 = 8$.